Google Play Store Apps Analysis

Project Report

Exploratory analysis of 10,841 Google Play Store app listings with Python and pandas

1. Introduction

This project explores a dataset of Google Play Store apps. It checks the quality of the data, finds and removes a corrupted record, and answers questions about ratings, categories, reviews and the difference between free and paid apps.

2. Objectives

- Inspect the structure, data types and summary statistics of the dataset.
- Detect and remove data errors.
- Rank categories by average rating and count perfectly rated apps.
- Convert Reviews to numbers and find the most reviewed apps.
- Compare free and paid apps by count and average reviews.
- Search app titles by keyword.

3. Tools and Technologies

Tool	Purpose
Python	Programming language
pandas / NumPy	Loading, cleaning, filtering, grouping, aggregation
Jupyter Notebook	Interactive development environment

4. Dataset

The file *googleplaystore.csv* has **10,841 rows and 13 columns**: App, Category, Rating, Reviews, Size, Installs, Type, Price, Content Rating, Genres, Last Updated, Current Ver and Android Ver. Only Rating is numeric on load (9,367 non-null values); Reviews, Size, Installs and Price are stored as text.

Rating summary (before cleaning)

Statistic	Count	Mean	Std. dev.	Min	25%	Median	75%	Max
Rating	9,367	4.19	0.54	1.0	4.0	4.3	4.5	19.0

A maximum rating of 19 is impossible on a 1-5 scale. It comes from one corrupted record (row 10472) whose values are shifted by one column: its Category reads `1.9` and its Rating reads 19. This is why the notebook later removes that row.

5. Methodology

- **Inspect:** `head`, `tail`, `shape`, `dtypes`, `info`, `describe`.
- **Search:** `str.contains('Astrology', na=False)` on the App column.
- **Group:** `groupby('Category')['Rating'].mean()` sorted in descending order.
- **Clean:** drop row 10472, drop the Rating column for the reviews analysis, and convert Reviews with `pd.to_numeric`.
- **Rank:** `nlargest(10, 'Reviews')`.
- **Compare:** filter on Type (Free vs Paid) and compare average Reviews.

6. Findings

6.1 Ratings and categories

Measure	Result
Unique categories	34 reported (33 real categories plus the corrupted value 1.9)
Highest average rating by category	1.9 (19.0, a data error), then EVENTS (4.44)
Apps with a perfect 5.0 rating	274
Mean rating	4.19, with a median of 4.3

The first-ranked 'category' is just the corrupted row. After ignoring it, EVENTS has the best average rating at 4.44. Most apps are rated between 4.0 and 4.5. The 274 perfect ratings are likely to include apps with very few reviews, which the notebook does not check.

6.2 Free vs paid

Measure	Free	Paid
Number of apps	10,039	800
Average reviews	478,661	11,673

About 93% of apps are free. The average free app has roughly 41 times as many reviews as the average paid app, though the averages are pulled up by a few giants. The two groups add up to 10,839, so two rows have neither a Free nor a Paid type.

6.3 Most reviewed apps

Facebook has the most reviews, followed by WhatsApp Messenger, Instagram and Messenger. The overall average is about 444,153 reviews per app. The top-10 list contains repeated apps (Facebook 2 times, WhatsApp Messenger 3 times and Instagram 4 times), which shows that the dataset contains duplicate rows.

6.4 Keyword search

Three app titles contain 'Astrology': Horoscopes - Daily Zodiac Horoscope and Astrology (4.6, 161,143 reviews), Min Thein Kha BayDin Astrology (4.7, 2,225 reviews) and iHoroscope - 2018 Daily Horoscope and Astrology (4.5, 398,307 reviews). All are in the LIFESTYLE category.

7. Issues and Recommended Improvements

#	Issue	Recommended fix
1	The corrupted row (index 10472) is removed only in later cells, so the mean rating (4.19), category count (34) and category ranking are calculated with it included.	Clean once at the top: `data = data.drop(10472)`, then recalculate.
2	Duplicate rows are not removed, so apps appear several times in the top-10 list and in averages.	Apply `drop_duplicates()` (for example on App) right after loading.
3	The same four-step cleaning code is copy-pasted into four cells, using hard-coded row numbers (`10472`, `loc[2544]`).	Create one clean DataFrame and use `idxmax()` or `nlargest`.
4	The paid-app average line is labelled 'For free apps'.	Change the label to 'For paid apps'.
5	Size, Installs and Price are text (such as '19M', '10,000+'), so they cannot be analysed numerically.	Strip symbols and convert to numbers, e.g. install counts and price.
6	1,474 apps have no rating and this is not reported or handled; the 5.0 count may include apps with very few reviews.	Report missing values and filter by a minimum number of reviews.
7	Averages of heavily skewed data (reviews) are used without medians.	Report medians as well as means.

8. Conclusion

The analysis gives a first overview of the Google Play Store: most apps are free, ratings cluster around 4.0 to 4.5, and a handful of social apps such as Facebook, WhatsApp and Instagram hold the most reviews. It also shows how a single corrupted row can distort results (a maximum rating of 19 and a fake category), and how duplicates inflate rankings. Cleaning the data once at the start, removing duplicates and converting the text columns would make the results more accurate.

9. Future Work

- Remove duplicates and recompute every result on the clean data.
- Convert Installs, Size and Price to numbers and analyse installs by category.
- Compare ratings of free and paid apps and by content rating.
- Analyse Last Updated to see how recently apps are maintained.
- Visualise rating distribution and the number of apps per category.
- Explore the relationship between reviews, installs and rating.